

The 600-Cell Grand Unified Theory: A Geometric and Group-Theoretic Model of the Standard Model and Beyond

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Abstract

This paper presents a Grand Unified Theory (GUT) based on the 600-cell, a four-dimensional regular polytope composed of five 24-cells. Building on previous work involving the isomorphism of Standard Model particles with the 24-cell and its dual, this model extends the mapping to the 600-cell using the alternating subgroup A_4 and the symmetric group S_5 . The 600-cell framework accommodates all fermions and bosons of the Modified Standard Model (MSM), including chirality, spin, and matter-antimatter duality. The model also incorporates the eight missing bosons proposed in the I Ching Leptogenesis framework, suggesting a deeper connection between symmetry, geometry, and particle physics.

1 Introduction

The Standard Model (SM) of particle physics is one of the most successful scientific frameworks ever developed. It elegantly describes the electromagnetic, weak, and strong nuclear forces through the gauge groups $U(1)$, $SU(2)$, and $SU(3)$, respectively. Despite its predictive power and experimental validation, the SM is incomplete. It does not incorporate gravity, explain the origin of mass for neutrinos, or account for the observed matter–antimatter asymmetry in the universe. These limitations have motivated the search for a more fundamental theory—a Grand Unified Theory (GUT)—that unifies all known interactions within a single, coherent framework.

Traditional GUTs, such as those based on $SU(5)$ or $SO(10)$, have made significant strides in unifying the SM gauge groups [11, ?]. However, they often fall short in addressing chirality, spin, and the full spectrum of particle generations. Moreover, they typically rely on abstract algebraic structures without a clear geometric interpretation. In contrast, the approach presented in this paper is rooted in the rich geometry of four-dimensional polytopes—specifically, the 600-cell—and their deep connections to group theory.

The 600-cell is a 4-dimensional polytope with 120 vertices, 720 edges and 1200 faces. Useful and interesting resources on the 600-cell can be found in [6, 4]. The geometry of the 600-cell is related to quasicrystals, a fact which has made it an alluring area of research for establishing a Theory of Everything. Several attempts in this direction have been made [7, 8, 9]. These attempts also focus on a method of creating the 600-cell from cuts and projections of the 8-dimensional polytope E_8 . Other examples of cut-and-project methods on quasicrystals and Jordan algebras can be found at [5].

The significance of E_8 and other exceptional Lie Groups for quantum mechanics has been established in a number of areas including String Theory and triality [13, 14]. The particles of the Standard Model have been applied to vertices of E_8 by Lisi [12]. This ambitious project set about applying all of the $U(1)$, $SU(2)$, and $SU(3)$ groups to the Exceptional Lie algebra and would include all matter/anti-matter fermions, their spins, chiral characteristics and colour charges embedded in multiple copies of $SL(2, \mathbb{C})$, which are known to exist in E_8 . However, there are not enough vertices in E_8 to accommodate all of these particles and their characteristics. At the present time, Lisi is looking at infinite groups to assign the particles to, which is a much more involved and difficult task.

In this paper, the author will attempt to show that it is entirely possible to apply all of the matter/anti-matter fermions of the Standard Model including their spins, chirality and colour to the vertices of the 600-cell, along with the electroweak, strong force and Higgs bosons. The model also includes 8 hypothetical I Ching Leptogenesis particles; six hexawens and two triwens, which are proposed as bosonic particles with fermion-like properties (e.g., chirality, hypercharge). These particles exhibit mass and charge oscillation and include negative mass states. The 600-cell only has 120 vertices, but it can be shown to accommodate all of these characteristics and more. Since E_8 is twice as large as the 600-cell, it is not outside the realms of possibility that a quasi-dual representation of the 600-cell could be used to create an isomorphism with E_8 . This duplicate 600-cell would encompass all of the Super Symmetric particles making E_8 a Super Symmetric particle model.

This work builds upon a sequence of prior investigations: - In O'Neill (2023a), the 24-cell was shown to be isomorphic to the binary octahedral group and $GL(2,3)$, providing a natural embedding for the particles of a Modified Standard Model (MSM). - In O'Neill (2023b), the mapping was extended to the 600-cell, which contains five 24-cells and 120 vertices. This allowed for a more complete representation of the MSM, including spin-up and spin-down states, and introduced the use of the alternating subgroup A_4 and the symmetric group S_5 to organize particle embeddings. - In O'Neill (2024), the I Ching Leptogenesis model introduced eight new bosonic particles—six hexawens and two triwens—to explain lepton number violation and matter-antimatter asymmetry.

The 600-cell GUT proposed here synthesizes these developments into a unified geometric and algebraic framework. By leveraging the symmetries of the 600-cell and its relationship to the groups A_4 , S_4 , and S_5 , the model provides a natural setting for embedding all known particles and interactions. It also offers a novel interpretation of symmetry breaking, chirality, and the absence of certain quantum numbers (such as hypercharge 5) through the lens of higher-dimensional geometry.

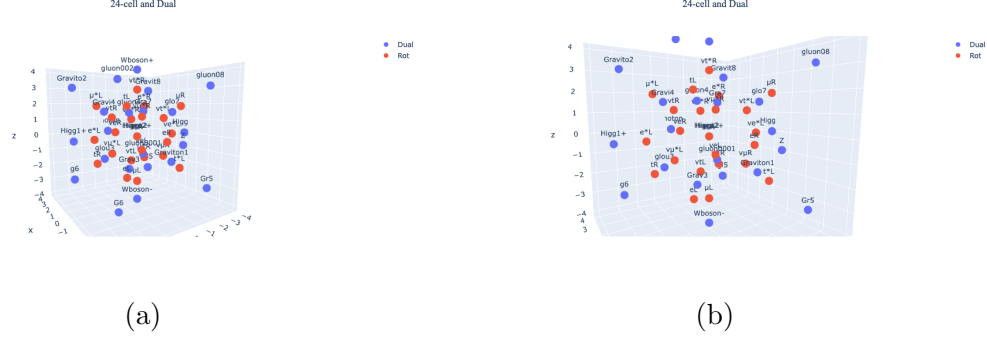


Figure 2: The fermion particles in the 24-cell dual are rotated by 0.25 radians in the zw and xy planes to align with the boson sector coordinates and thereby assign elements of the S_4 symmetric group to the fermion particles (Figure 2a). A close up of the same rotation (Figure 2b).

recognise that an automorphism between S_4 and S_5 exists. There are 24 elements in S_4 and 120 in S_5 . This means that there are 5 copies of S_4 in S_5 , which is equal to the number of 24-cell copies that exist in the 600-cell. Therefore, we can duplicate our 24-cell S_4 conjugacy isomorphism five times. This can be done using the exotic exclusion map $S_4 \rightarrow S_5$, which fixes one element in S_5 and permutes the remaining elements by S_4 .

The resulting system allows us to ascribe spin up and spin down characteristics to each of these S_4 subgroups (see Table in supplementary resource). The bosonic sector 24-cell is given all vertices that have 1 as their first S_5 element index value. The spin up lepton sector is given all S_5 elements with 2 as its first index. The spin down lepton sector is given all S_5 elements with 3 as its first index. The spin up quark sector is given all S_5 elements with 4 as its first index. And finally, the spin down quark sector is given all S_5 elements with 5 as its first index.

The cardinality of conjugacy groups S_5 are; $\{1, 10, 15, 20, 20, 24, 30\}$. The 600-cell has a latitude cell structure of $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, which is once again a rearrangement of the S_5 conjugacy groups and indexed 1-9. Each of the numbers in the 600-cell latitude cell structure (with the exception of the first and the last, which are the identity and inverse, respectively) represent a polytope with an equivalent number of vertices; the sets of 12 vertices comprise a dodecahedron, the 20 is an icosahedron and the 30 is an icosidodecahedron. It is a simple matter to ascribe all of the S_5 4-cycle conjugacy groups to the icosidodecahedral vertices. However; how do we know, for example, which set of twenty vertices from the S_5 conjugacy group map to which set of icosahedrons in the 600-cell? And if we are to question that, then there are a whole host of other questions of a similar kind that we could ask. For example, which half of the 24 disjoint cycles map to which dodecahedron? And are they really halved in a neat manner? Perhaps they are interleaved or completely random assignments. For that matter, perhaps all of the conjugacy groups are internally randomly assigned.

Therefore, a much better option is to use the even permutations of the 600-cell vertex generators and using the A_4 subgroup. The eight permutations of

$$\left[\frac{\phi}{2}, \frac{1}{2}, \frac{\phi^{-1}}{2}, 0 \right] \quad (1)$$

times the even permutations of A_4 makes 96 vertices. Half of these A_4 sets can be converted into the remaining S_4 elements by multiplication with a transposition element of S_4 : $f \circ g$, where $f \in A_4$, $g \in S_4 \setminus A_4$. This implies that there are 12 such possible combinations. This multiplied by the original 12 combinations of Clifford Rotations of the 24-cell makes for a total of 144 possible combinations. This is still a lot more accurate than the over 10,000 possible Calabi-Yau surfaces proposed in String Theory. However, it is possible to assign a set of A_4 times an element in $S_4 \setminus A_4$ permutations to each set of vertices under rotation, which means the 144 combinations can be reduced to just 12 again. Therefore, each particle would be a 1/12 choice (equivalent to -1/12 or infinity?). Having multiple vertices and therefore multiple matrices for each equation makes sense, as each particle needs to be multiplied by the four Pauli matrices (which are a subgroup of $SL(2, \mathbb{C})$ and these should be arrived at from these 12 matrices, or by one other matrix from $SL(2, \mathbb{C})$. Exactly which matrix this is might reveal further interesting clues, but it is beyond the scope of this paper and this research, for now.

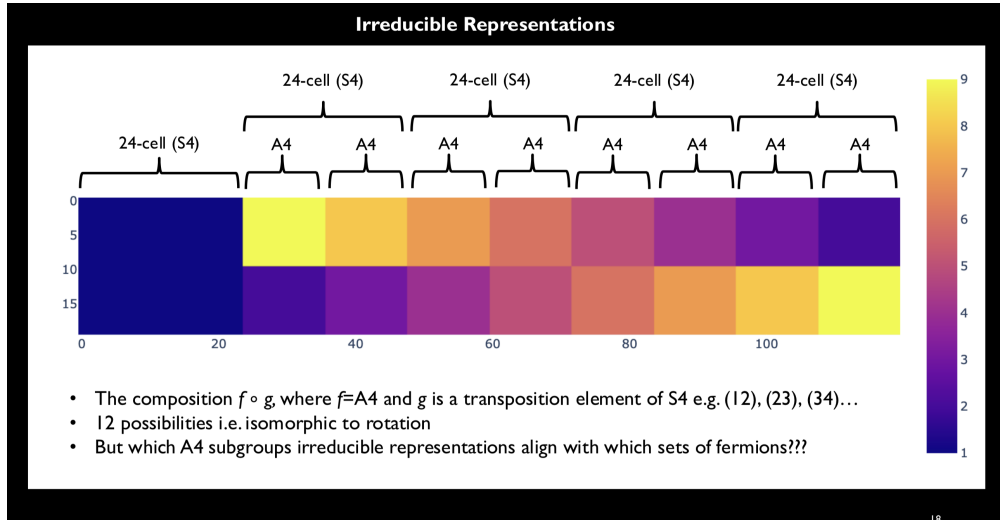


Figure 3: Irreducible representations of the 600-cell.

For the sake of expediency and simplicities sake, the alternating groups of A_4 are multiplied by the S_4 transposition 2-cycle (12). This provides the remaining S_4 isomorphisms. But it does not tell us which A_4 groups need multiplication. To do this, we apply the same dual rotations to the 600-cell generators, as we did earlier with the 24-cell; namely, $(4 \times \pi/4) \times ((xw, yz), (yw, xz), (zw, xy))$. This results in Figure 3. The 600-cell is constructed from one 24-cell formed by combining the 8-cell and 16-cell vertices. The remaining 96 vertices are generated by even permutations of coordinate sets, forming eight copies of the A_4 group. By examining the top row in Figure 3 and comparing it with the bottom row, it is apparent that the S_4 subgroup corresponding to the one 24-cell – mentioned above – rotates into itself i.e. it is an irreducible representation. The remaining A_4 subgroups, which correspond to the remaining 96 vertices appear to be in reversed order. At the centre of this

group we see that the four subgroup crossover and switch places. Therefore, it is correct to assume that we read the graph downwards i.e. the irreducible representation of any A_4 subgroup in the top row can be found directly beneath it, in the bottom row.

By multiplying the paired A_4 subgroups in the bottom row with our transposition element (12), we can recover the S_4 groups and thereby assign our quarks and leptons to the 600-cell vertices. The final step is to add the spins and colour charges. The spin up and spin down characteristics have the following sequence (0, 1, -1, 1, -1), where zero spin is applied to the bosons and the rest are either spin up or down. By virtue of the irreducible representations, it should be impossible for the quarks and leptons to rotate into one another, which is important as it establishes the stability of the proton; something which the SU(5) GUT was shown to be incapable of [11].



Figure 4: Weak isospin graphs of the $\bar{e}_L$, $\bar{e}_R$ and e_L particles, respectively.

The directed and bi-directional graphs of S_8 allow us to apply the electroweak model, where 1/2 particles are right-handed particles, -1/2 are lefthanded particles and the bi-directional graphs cancel to zero, meaning that they do not interact with the Weak Force. Therefore, we can apply the Weak isospin (I_3) for these fermions, in this manner, and from there apply the hypercharge ($Q - I_3$). This addition gives the current proposed 600-cell model the equivalent features seen in the Georgi–Glashow model (or SU(5) GUT) [11]. While the vertices of the 600-cell are 5-colourable, nevertheless, the vertices of the embedded 24-cells are 3-colourable. Therefore, the fundamental representation of SU(3) color charge may still be applicable in this system. Permuting these three colours allows us to embed all 288 fermion classes (matter/anti-matter, chirality, spin, iso-spin, hyper charge and colour charges) into just 96 vertices of the 600-cell.

As a consequence, this system implies that the quark colour charges are not an explicit characteristic of the model, but are rather dynamic in some way. This undoubtedly true in the nucleus itself, where colour charge is a dynamical process. But from the perspective of the model, it appears to suggest that talk of a ‘red quark’ or a ‘blue quark’ is not actual or intrinsic. They are not real things. The quarks are real, but the charges are just something that they can possess or not possess at any given moment. The colour charges are not relevant to the bosons or leptons, but are present in any case from a symmetry point of view.

When the relations to the embedded 24-cell model are brought into consideration, this model includes; the U(1) electromagnetic force, SU(2) fermion spin, chirality and spin statistics, the fundamental representation of QCD, Electro-weak symmetry breaking, a model of the Weak Force, 8 hypothetical I Ching particles (needed for leptogenesis) and a canonical graviton that is equidistant to all particles and therefore interacts with all of them equally – making this model the most complete and self-consistent model of all of the fundamental

forces and the particles of the Standard Model and beyond.

Each of the five 24-cells embedded in the 600-cell is assigned to a distinct particle sector: one for bosons, two for leptons (spin-up and spin-down), and two for quarks (spin-up and spin-down). The assignment of particles to vertices is governed by the irreducible representations of the group S_5 , which has 120 elements—matching the number of vertices in the 600-cell. The conjugacy class structure of S_5 is mapped to the latitudinal structure of the 600-cell, with each class associated with a specific polyhedral structure.

The eight remaining unassigned vertices in the bosonic 24-cell are occupied by the hexawen and triwen particles introduced in the I Ching Leptogenesis model. These particles are bosonic but exhibit fermion-like properties and their interactions violate lepton number conservation—providing a mechanism for the observed matter-antimatter asymmetry.

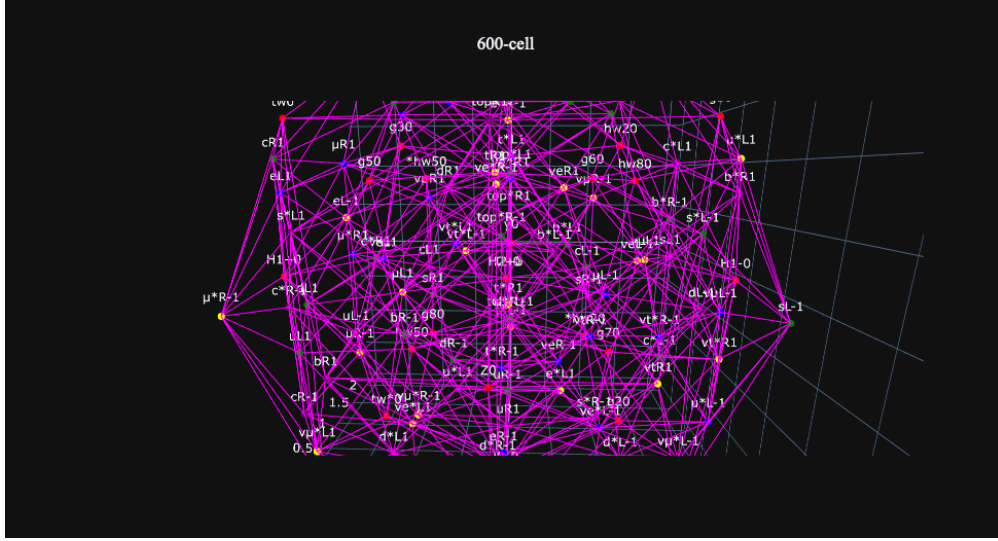


Figure 5: Particle assignments across the 600-cell.

3 Discussion

The 600-cell Grand Unified Theory (GUT) presents a compelling synthesis of geometry, group theory, and particle physics. By embedding the Modified Standard Model (MSM) within the structure of the 600-cell, this model offers a novel and highly structured approach to unification.

One of the most striking features of the 600-cell GUT is its ability to encode complex particle properties—such as spin, chirality, and matter-antimatter duality—within a purely geometric structure. The use of five embedded 24-cells allows for a natural partitioning of the particle spectrum into bosons, leptons, and quarks, each with spin-up and spin-down states. This geometric encoding is grounded in the symmetry properties of the 600-cell and its isomorphisms with the binary icosahedral group, $SL(2,5)$, and the symmetric group S_5 .

The use of irreducible representations (irreps) of S_5 to assign particles to the 600-cell's vertices introduces a powerful mechanism for organizing the particle spectrum. These irreps

correspond to distinct polyhedral structures within the 600-cell and their alignment with particle types suggests a deeper connection between symmetry and physical identity.

The integration of the I Ching Leptogenesis framework enhances the model's explanatory power. The hexawen and triwen particles, while bosonic in nature, exhibit fermion-like properties and play a crucial role in lepton number violation. Their annihilation in particle simulations aligns with the expected behavior of leptogenesis, and their placement within the 600-cell reinforces the model's internal consistency.

Despite its elegance, the 600-cell GUT raises several important questions. While the model is mathematically rich, its predictions must ultimately be testable. However, calls for the detection of hexawen and triwen particles would be premature – since all such particles were eradicated almost as soon as they were created. It has been demonstrated that indirect evidence of signatures of their interactions have been found in the CMBR power spectrum fluctuations – indicating that the CMBR is not the remnant of the Big Bang, but rather the echo of the war between matter and anti-matter in the leptogenesis event. However, the use of multiple symmetry groups in the 600-cell introduces significant complexity, and ensuring that these mappings are not redundant or over-constrained will require further formal analysis. Whether this model can be naturally extended to E_8 also remains an open avenue for exploration.

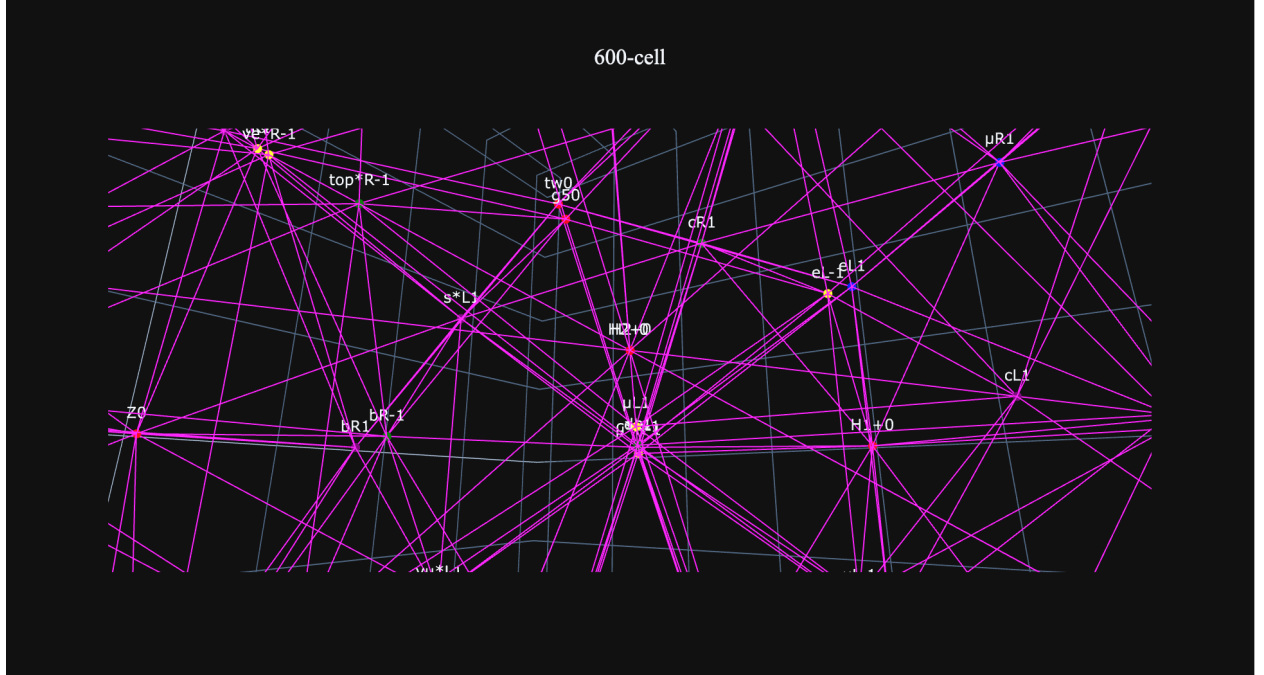


Figure 6: Close up of the particle assignments across the 600-cell.

Some interesting asymmetries arise in the mapping of the fermions to the I Ching-gluon sector. Only 3 right-handed and 5 left-handed chiral S8 permutations are assigned to the gluons. The situation is reversed with the I Ching particles, naturally. Moreover, only 3 matter and 5 anti-matter coordinates are assigned to the gluons, with 5 and 3 being assigned to the I Ching particles, respectively. Remarkably, this lines up with proposed chirality of the I Ching particles, which possess 5 left-handed and 3 right-handed particles

$hw_L, hw_R, h\bar{w}_L, \hat{h}w_L, \hat{h}w_R, \hat{h}w_L, tw_R, tw_L$ – provided the anti-triwens are ignored. This is a stunning confirmation of both the I Ching Leptogenesis model and the 600-cell GUT, which also tells us that we have mixed up the vertices for the gluons and I Ching particles and that they need to be reversed, at some later stage.

Another potential asymmetry arises: There are 6 I Ching particles assigned to the right-handed S_8 permutations and only 2 assigned to lefthand matrices. The same is also true of the gluons, however in the case of the gluons it is 6 left-handed and 2 right-handed S_8 matrices. This is interesting when we consider that 2 of the Gell-Mann matrices are constructed by linear additions of other matrices. This suggests that the equivalent set of matrices for the I Ching particles (if any such one exists) would have 6 linear combination matrices and only 2 pure (or fundamental) matrices. However, since we have already established that the gluon and I Ching sectors are out of place, this may suggest that the SU(3) structure is more fundamental to the I Ching particles and only borrowed (or ported over) to the gluons.

In an over-arching sense, the main insights of this research can be summarised as: The fact that spin-1/2 particles, QCD, and many of the other aforementioned attributes of the Standard Model can be comprehensively modelled by 4D spatial geometry suggests that such systems are inherently 4-dimensional, by nature.

4 Future Work

Just as there are five copies of the 24-cell in 600-cell, there are five copies of the 600-cell in its dual, the 120-cell. This implies that we can duplicate the vertices of our current model to the vertices of the 120-cell. These 5 duplicates could relate to the 5 different types of String Theory, although this is highly speculative. Alternatively, we can stay working with the 600-cell, but use the dual cells of the 120-cell as dodecahedral particle wells for use in Lagrangians or Path Integrals. The 120-cell has some interesting helical properties which may be worth exploring, from this point of view – as well as from the perspective of genetics and therefore to abiogenesis. This work is further a stepping stone to compactifying the dimensions of the 120-cell to just three dimensions, which could help produce useful techniques, or else merely reinforce (add more credence to) the 120-cell lattice method.

5 Conclusion

All 288 fermion classes (matter/anti-matter, chirality, spin, and colour charges) along with 24 bosons were ascribed vertices in 600-cell. When the relations to the embedded 24-cell model are brought into consideration, this model includes; the U(1) electromagnetic force, SU(2) fermion spin, chirality, Lorentz transforms, and spin statistics, the SU(3) fundamental representation, Electro-weak symmetry breaking, and a model of the Weak Force. This makes the 600-cell GUT the most complete and self-consistent model of all of the fundamental forces and the particles of the Standard Model and beyond.

The 600-cell Grand Unified Theory (GUT) represents a bold and imaginative step toward reconciling the fragmented domains of particle physics, geometry, and group theory. By

embedding the Modified Standard Model (MSM) within the intricate structure of the 600-cell, this framework offers a fresh and compelling vision of unification.

At its core, the model demonstrates that the full spectrum of MSM particles, including their spin, chirality, and matter-antimatter duality, can be systematically mapped to the vertices of the 600-cell. This is achieved through the use of irreducible representations of the symmetric group S_5 , which mirrors the 600-cell's vertex count and internal structure.

One of the most innovative aspects of the 600-cell GUT is its integration of the I Ching Leptogenesis framework. The introduction of eight new bosonic particles—six hexawens and two triwens—addresses a longstanding gap in the 24-cell and 600-cell models. These particles play a crucial role in lepton number violation and offer a plausible mechanism for matter-antimatter asymmetry.

While the model is rich in structure and insight, several challenges remain. Theoretical elegance must be matched by experimental evidence. The model's reliance on higher-dimensional geometry and abstract group theory must be reconciled with observable physics. Nonetheless, the 600-cell GUT stands as a testament to the power of interdisciplinary thinking and the enduring relevance of symmetry in the quest to understand the fundamental nature of reality.

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